

EXPLORING HIGH-DOSE-RATE ELECTRON IRRADIATION FOR RADIOBIOLOGY USING A 10 PW LASER SOURCE

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Abstract. Advancements in high-power laser technology have opened new possibilities for electron acceleration, enabling the development of novel irradiation platforms for radiobiology studies. Ultrahigh dose rate (UHDR) radiation delivery is emerging as a promising approach for cancer therapy. The FLASH effect, observed under UHDR irradiation, has demonstrated the ability to preserve antitumor efficacy while reducing damage to healthy tissues. High-intensity petawatt laser systems (such as the one available at ELI-NP) offer a potential solution, capable of generating very high-energy electrons (VHEEs) at extreme dose rates greater than to 10^{12} Gy/s in ultra-short pulses (<1 ps).

This paper reports for the first time the design, implementation, and characterization of a setup dedicated to the irradiation of live cells (healthy and cancerous) using electrons produced by the interaction of a 10 PW laser with a mixed gas jet target ($N_2 + He$). The experimental setup was optimized to ensure precise evaluation of electron beam parameters like energy distribution, dose rate, and spatial uniformity, making it suitable for controlled biological experiments. Dosimetry was performed using radiochromic films (RCF), ionization chambers (IC), and Optically Stimulated Luminescence detectors (OSL), ensuring accurate dose delivery to cell cultures. To validate the system, *in vitro* assays were conducted to assess cell viability post-irradiation, confirming the feasibility of using such an electron source for radiobiological studies. This work provides a foundation for further exploration of ultrahigh-dose-rate irradiation and its potential applications in radiotherapy research.

Key words: ultrahigh dose rate (UHDR), laser-driven electron radiation, radiation therapy.

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1. INTRODUCTION

Recent advancements in high-intensity laser systems have enabled the generation of ultrashort, high-energy electron beams through laser-plasma interactions, offering new avenues for radiobiological studies. Compared to conventional irradiation methods based on particle accelerators, laser-driven sources provide unique beam characteristics, including high peak dose rates, short pulse durations, and potential applications in FLASH radiotherapy [1–4].

The motivation for developing a dedicated setup for irradiation of live cells using a high-power laser, stems from the unique features of the laser-accelerated particle beams, as for instance instant high-dose and high-dose-rate, and the versatility in changing the particle type (*e.g.*, electron, ion). Preliminary studies indicate that FLASH radiotherapy [5] may limit damage to normal tissues while maintaining tumor control. Investigations with electrons [6, 7], protons [8], and photons [9] suggest a link between this protective effect and the radiation-induced oxygen depletion. In healthy tissues, oxygen plays a critical role in stabilizing DNA damage, whereas tumors often exist in hypoxic environments, which may contribute to their differential response [10]. However, this hypothesis has been recently questioned [11], emphasizing the need for a more detailed understanding of dose-rate effects before clinical adoption.

While conventional FLASH irradiation delivers dose rates on the order of tens to hundreds of Gy/s, recent research is exploring even higher rates [4, 8, 12] to investigate the underlying biological mechanisms of tissue sparing and to identify the thresholds at which these effects may occur or plateau.

High-power laser systems are suitable for such investigations, owing to the intrinsic generation of ultrashort radiation pulses ranging from tens of femtoseconds to tens of nanoseconds [13]. Laser-driven sources have been employed in initial radiobiological studies exploiting both low- and high-LET particles, such as electrons, photons, protons, and ions. Nonetheless, practical studies remain limited due to the challenges of delivering precise and reproducible doses on extremely short timescales [12].

This study reports the development and characterization of an irradiation setup designed for radiobiological experiments using electrons generated by a 10 PW laser-driven gas target. This work focuses on the technical aspects of the experimental setup, including beam characterization, dosimetry validation, and the design of the irradiation chamber.

To verify the system functionality biological assays (cell viability and micronuclei formation) were included without being the primary focus of the present study. The goal is to establish a reliable platform for investigating the radiobiological effects of laser-accelerated particle beams, and to contribute to the development of future high-dose-rate radiotherapy applications with different types of particles (*e.g.* electron, ion) as for instance the project “Medical Applications of High-Power Lasers–Dr. LASER,” of ELI-NP [14].

2. MATERIALS AND METHODS

The experimental setup was carried out using the 10 PW laser system, and the long focal geometry available in the E6 experimental target area of ELI-NP [16]. The high-power laser system (HPLS) is a hybrid Ti-Sa system with a central wavelength of 810 nm. The laser delivered pulses with a typical full width at half maximum (FWHM) duration of 23 fs, a maximum on-target pulse energy of up to 230 J at a repetition rate of 1 pulse per minute (1/60 Hz) [15–20]. The beam is focused by a spherical mirror over ~ 30 m [21], producing a focal spot with a diameter of 58 ± 2 μm at FWHM, with an encircled energy of about 50%, giving an effective laser peak intensity on target of $\sim 2 \cdot 10^{20}$ W/cm².

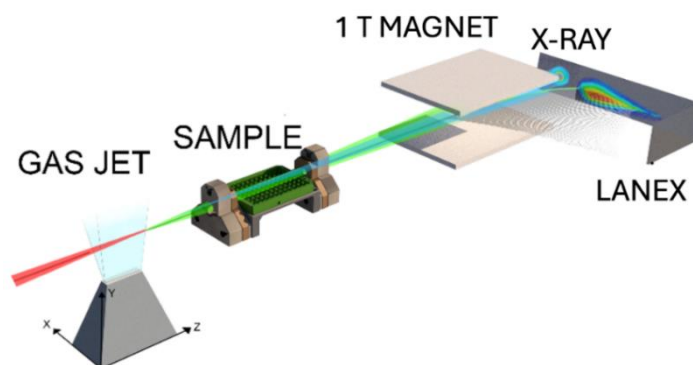


Fig. 1 – Sketch of the experimental set-up. The laser beam is focused onto a gas jet target to produce relativistic electrons *via* the LWFA mechanism, which interact with the biological sample (live cells). The electron spectra are evaluated by means of a multi-GeV electron spectrometer, and most of the laser parameters are measured on-shot.

The experimental configuration is shown in Fig. 1. The laser beam is focused on the gas jet target, generating a highly energetic beam of electrons *via* the Laser Wakefield Acceleration (LWFA) mechanism. The electron beam produced in the interaction chamber under 10^{-6} mbar vacuum exits the chamber through a (250 μm) thin Kapton window and interacts with the biological sample, and its spectrum is evaluated using a spectrometer consisting of a 1 T dipole magnet and a scintillating LANEX screen.

2.1. BIO SAMPLE DESIGN AND DOSIMETRY

To expose the cell lines to the laser-driven radiation, an irradiation setup was developed, as illustrated in Fig. 2. The samples were placed in 0.5 ml Eppendorf

tubes on the sample holder. Accurate dose monitoring within the cell-irradiation platform is ensured by a dedicated dosimetry system integrated along the laser-driven beamline. The system consists of two sets of combined active and passive detectors, positioned before and after the irradiation region of the cell-irradiation setup, with respect to the beam propagation direction (see Fig. 2). The active dosimeters consist of plane-parallel ionization chambers (Advanced Markus, PTW Type 34045), while the passive dosimetry is provided by optically stimulated luminescence (OSL) detectors (type BeOSL, based on beryllium oxide chips) and film detectors (radiochromic films EBT3). The close spatial placement of active and passive dosimeters enables direct cross-validation of local dose measurements, thus improving the reliability and robustness of the overall dose determination. By comparing upstream and downstream measurements, the effective dose delivered to the biological samples can be accurately inferred. Such a comprehensive configuration is essential for quantitative radiobiological experiments and for performing systematic dose–response studies.

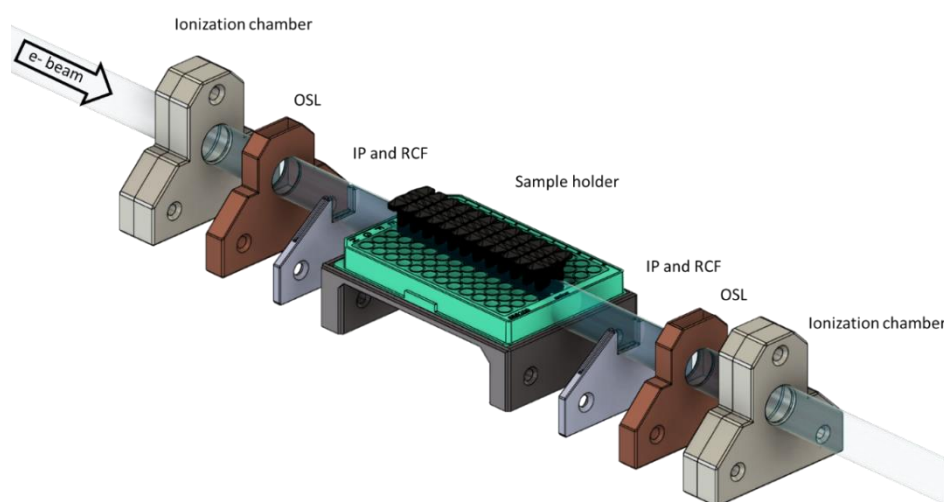


Fig. 2 – The design of the irradiation setup. The system is composed of the passive and online radiation detectors: ionization chamber, OSL, RCF, IP, and the sample holder.

The active dosimetry detectors, PTW Advanced Markus ionization chambers, are specifically designed for high-dose-rate particle beams [22], which could recommend them for laser-driven applications. Each chamber features a nominal sensitive volume of 20 mm³, a collecting electrode diameter of 5 mm, and an entrance window with a thickness of approximately 0.03 mm PE (polyethylene CH₂), minimizing beam perturbation while maintaining high sensitivity. The electrode

spacing of 1 mm ensures fast charge collection and reduced charge recombination, which is critical under the high instantaneous dose rates characteristic of laser-based radiation sources.

Having a weak energy dependence (in the relevant X-ray range) and a compact form factor, the BeOSL detectors are characterized by high sensitivity, a near tissue-equivalent effective atomic number ($Z_{\text{eff}} \approx 7.1$), and a broad dynamic dose range spanning from the μGy level up to several Gy. The recorded signal is subsequently processed by an automated reader to derive the operational quantity $H^*(10)$, as described by Jahn and collaborators [23]. The dosimeters are calibrated by the manufacturer (Mirion/Dosimetrics) [24] at the Helmholtz Zentrum München (HMGU), with traceability to the primary standard Physikalisch-Technische Bundesanstalt (PTB), Germany. Additional periodic in-house verifications are performed using encapsulated Sr–Y sources in a dedicated irradiator [24], with acceptance criteria of deviations below 8%. The BeOSL reader is traceable to conventional continuous radiation fields; consequently, its calibration was carried out for 4.396 mSv (in $H^*(10)$ quantity), corresponding to 4mGy air kerma, in a Secondary Standard Dosimetry Laboratory using a S-Cs reference source. The results of the ISO 62387 linearity assessment indicate a relative standard deviation of 5.56% within the tested dose range between 0.05 mSv and 1 Sv, for Co-60. For this studied dose range, $H^*(10)$ to Air Kerma factor was calculated as 1.1418 ± 0.11 . Recent investigations carried out at the ALFA multi-MeV laser–plasma electron accelerator have demonstrated favorable performance for certain passive dosimetry systems. In this context, Cimmino and collaborators [25] reported that BeO-based dosimeters are appropriate for use in petawatt-class laser-driven experiments at ELI Beamlines, which are comparable to those conducted at ELI-NP.

Radiochromic film dosimetry was employed as an independent, passive method for validation of the absorbed dose delivered to the biological samples. Gafchromic™ EBT3 films were used for measurements in the Gy range due to their near tissue-equivalent effective atomic number, weak energy dependence for MV photon and high-energy electron beams, and high spatial resolution [26]. The films were positioned on the beam axis at two locations: immediately upstream of the sample to quantify the incident dose, and downstream of the sample to assess transmitted dose and potential attenuation effects. After irradiation, the films were scanned under controlled conditions and analyzed using calibrated dose–response curves [27, 28], enabling cross-validation of the active dosimetry measurements and assessment of spatial dose uniformity along the beam axis.

Among the various schemes developed for laser-based electron acceleration, LWFA is currently the most widely employed. Originally proposed by Tajima and Dawson in 1979 [29], LWFA has evolved into a mature and powerful technique for producing high-energy electron beams over the past decades. Its effectiveness relies on the extremely strong electric fields generated by plasma waves driven by ultra-intense

laser pulses, with accelerating gradients that can reach hundreds of GV/m-far exceeding those achievable with conventional radio-frequency accelerator technology [30]. This capability enables electron acceleration to GeV energies over centimeter-scale distances, paving the way for compact accelerator systems and a wide range of applications requiring high-energy particle sources. The achievable acceleration length and final electron energy are governed by the plasma density and laser parameters, while the electron trapping process depends critically on the injection mechanism, such as self-injection, ionization injection, or external injection [29].

2.2. IN VITRO ELECTRON IRRADIATION

Cell line. Murine fibroblast cell line, L929, was purchased from ATCC (Virginia, USA) and cultured in Dulbecco's modified Eagle medium (DMEM) supplemented with 10% fetal bovine serum (FBS) and penicillin–streptomycin (0.5%–100 units/mL). For the irradiation experiments, the cells were detached 30 min before irradiation and placed in PCR tubes at a density of 100.000 cells per tube and placed in the experimental set-up described in Fig. 2. Control cells were manipulated similarly, without being exposed to radiation. Following irradiation, the cells were diluted to the desired density and further grown in 24 well plates. All cell cultivation media and reagents were purchased from Biochrome AG (Berlin, Germany).

Cell viability assay. Cell viability was assessed using the MTT assay (Serva, Heidelberg, Germany) at 24 post irradiations. First the medium was removed, 1 mg/mL MTT solution was added to each well and the cell culture was further incubated for 4 h. Then the medium was removed and the formazan crystals were dissolved by DMSO (Merck, Darmstadt, Germany). Optical absorbance was recorded at $\lambda = 490$ nm using a Mithras LB 940 plate reader (Berthold, Germany). Cell viability was normalized to control cells.

Micronuclei (MN) assay. After irradiation, the cells were plated at a density of 10.000 and incubated at 37°C for 24 h and Cytochalasin B (6 μ g/mL) was added 4 h after culture initiation. Cells were then harvested and fixed and colored with Acridine-Orange. For each sample, 500 binucleated cells were scored under an optical microscope (final magnification 40 $\times$) for micronucleus analysis, following the criteria for micronucleus acceptance [31]. MN frequency was evaluated and reported to the control samples.

Statistical analysis. Data are presented as the mean $\pm$ standard deviation (SD) of at least three independent experiments with three replicas per experiment, if not specified otherwise. Statistical analysis was performed using GraphPad Prism 5.01 software (San-Diego, CA, USA). The statistical significance of differences between experimental groups was calculated using Two-way analysis of variance with Tukey's Multiple Comparison Test. The values of $p < 0.05$ were considered statistically significant.

3. RESULTS AND DISCUSSIONS

3.1. PARTICLE BEAM CHARACTERIZATION AND DOSIMETRY

To characterize the electron beam generated by the laser–plasma interaction, a series of radiochromic films (RCFs) were positioned at multiple locations along the beam propagation axis. These detectors provided spatially resolved measurements of the electron beam profile, enabling the estimation of beam divergence and dosimetry. Analysis of the collected data indicates that the electron beam has a diameter at the sample location of ~ 1.4 cm, exhibiting a divergence of approximately 3 mrad FWHM. Figure 3a shows an imprint of the electron beam on the RCF which is placed directly in front of the biological sample to be subsequently irradiated, providing a direct visualization of the beam profile and at the same time, the estimation of the dose received by the biological sample.

The electron energy spectrum was measured on a shot-to-shot basis using a LANEX scintillating screen placed downstream of a 1 T dipole magnet. The magnetic spectrometer deflected downwards the electrons according to their energy, allowing reconstruction of the spectrum from the position-resolved LANEX signal. The spectrum shown in Fig. 3c presents the collected spectrum of the electron beam obtained from a shot on the LANEX without the sample. It consistently exhibits a broad profile with a pronounced peak around 2 GeV, characteristic of laser wakefield acceleration. A first estimation of the charge of each electron bunch can be determined using the spectrometer system, obtaining an average value of 0.4 nC per pulse (see Fig. 3b). Over 14 accumulated shots, this corresponds to a total delivered charge of approximately 5 nC. However, the electron spectrometer has a lower cut-off at approximately 1 GeV electron energy, therefore clearly underestimating the actual charge that hit the bio sample. Based on the formalism provided in [26], and using the measured energy spectrum of each shot along with the collisional stopping power of relativistic electrons in water-equivalent media, the expected absorbed dose for the full irradiation sequence was calculated to be approximately 6.4 Gy. The main purpose of the electron spectrometer is, in fact, to help find stable conditions for the interaction and to get an approximate evaluation of the dose per shot. Although the experiments were performed using the 10 PW laser system at ELI-NP, the laser was not operated at its nominal maximum energy. The effective on-target power during the irradiation campaign was approximately 8 PW. In addition to the reduced laser energy, variations in gas density directly affect the plasma wakefield structure, injection dynamics, and beam loading, thereby impacting the total accelerated charge and spectrum.

Accurate dose estimation was performed using EBT3 radiochromic films positioned along the beam axis. The films indicated an accumulated dose of approximately 10 Gy (see Fig. 3a), averaged over the entire imprint of the beam, corresponding to ~ 0.7 Gy per pulse. The higher dose measured by film compared

with the LANEX-based estimate is consistent with the contribution of sub-GeV electrons that are excluded from the LANEX field of view due to deflection and transport losses. These lower-energy electrons, although not fully represented in the charge estimate by the LANEX, contribute significantly to energy deposition in millimeter-scale biological samples because of their higher stopping power and perhaps high flux.

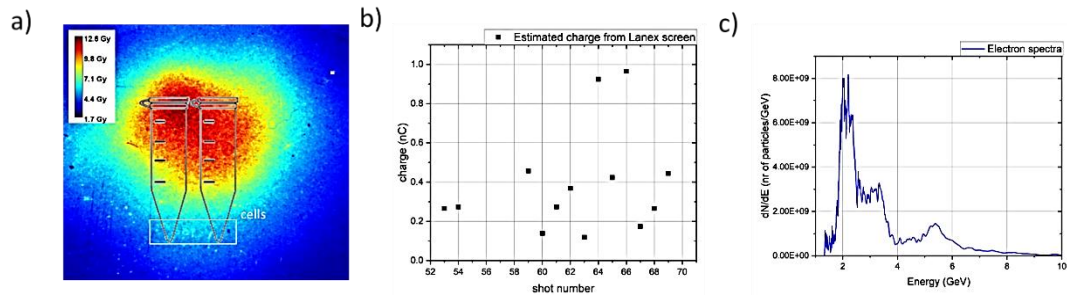


Fig. 3 – a) Dose map obtained from the RCF film positioned in front of the sample. The outlines of the two sample-containing tubes are superimposed to indicate the position and extent of the irradiated region relative to the measured beam profile. b) Charge distribution per shot estimated from the LANEX screen. c) Example of electron spectrum obtained from the interaction between the high-power laser and a 60 mm gas-jet He target with 2% N₂ as dopant.

Online monitoring was performed using an Advanced Markus 34045 plane-parallel ionization chamber operated at 200 V. The chamber recorded an average uncorrected dose of 0.16 Gy per pulse. This dose is significantly lower than the one recorded by the RCF, indicating that the gas ionization and recombination that occur under ultra-high dose rate conditions are greatly affecting the measurement. In the ultra-high dose-per-pulse regime investigated here, the response of the plane-parallel ionization chamber is significantly affected by ion recombination. According to Boag's theory for pulsed radiation beams [32, 33], general (volume) recombination increases with dose per pulse and charge density in the sensitive volume, leading to a systematic underestimation of the absorbed dose if no correction is applied. Therefore, appropriate recombination correction factors must be considered when ionization chambers are used for online dosimetry in laser-driven irradiation conditions. Based on the experimentally observed discrepancy between the radiochromic film dose (0.7 Gy per pulse) and the uncorrected Advanced Markus chamber reading (0.16 Gy per pulse), an empirical ion recombination correction factor of approximately 4.4 was determined. This correction factor is consistent with values reported in the literature [34–36] for plane-parallel ionization chambers operated in an ultra-short pulsed radiation regime. By applying this correction factor, the doses obtained with the Markus chamber are consistently scaled to the film measurements within our experimental uncertainties. Given a dose of approximately 0.7 Gy per pulse and an estimated electron bunch duration of about a hundred fs (based on our calculations), the instantaneous dose rate is evaluated to $\sim 10^{12} - 10^{13} \text{ Gy} \cdot \text{s}^{-1}$, similarly with data presented in literature [13].

Complementary offline measurements using BeOSL dosimeters yielded a cumulative dose of 4 Gy Air Kerma, which lies between the 2.2 Gy measured by the Advanced Markus ionization chamber and the higher values obtained from radiochromic film (~ 10 Gy), further highlighting the limitation of active detection and the importance of cross-validating the absolute dose in ultra-high dose-rate, laser-driven conditions.

3.2. CELL VIABILITY ASSAY- MTT AND MN ASSAY

Spatial analysis of the radiochromic film indicates that the biological samples were positioned on the lower-intensity side of the transverse beam profile. Consequently, although the maximum dose on axis reached approximately 10 Gy over 14 shots, the effective dose delivered to the cells is estimated to be approximately 4 Gy for the full irradiation sequence.

In Fig. 4A the results of the L929 and B16 cell viability evaluated at 72 h following electron irradiation at 4 Gy and ultra-high dose rate are presented. The data show that the viability of the melanoma cells decreases for the irradiated cells, to $74\% \pm 22$. On the contrary, for L929 cells, we see a slight increase following irradiation ($108\% \pm 19$). Previous studies have shown that metabolic activity can be evaluated effectively by the MTT assay [37–39]. While for the cancerous cell line, the viability decreases with increasing dose, indicating a loss of cell proliferation, the increase in cell viability for the normal cell line can be attributed to an increase in mitochondrial mass. This was also reported previously by Rai and collaborators for several cell lines [38]. Andreassi and collaborators also showed that laser-driven electron pulses inhibit the growth of the OVCAR-3 cancer cell line [40]. Similar results for melanoma cells were also reported after gamma radiation [41].

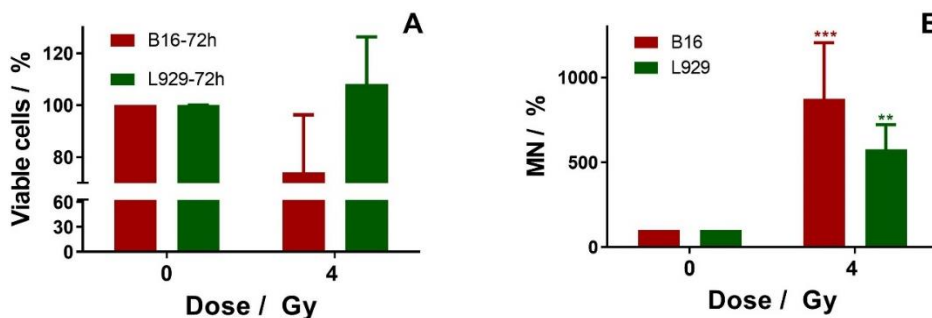


Fig. 4 – a) L929 and B16 cellular viability and b) Percentage of MN for L929 and B16 cells as a variation after irradiation with 4 Gy. Data are presented as mean $\pm$ SD. p values were calculated using ANOVA analysis with Tukey's multiple comparison post-test. ** $p < 0.01$, *** $p < 0.001$.

The percentage of MN was evaluated at 24 h post-irradiation, and the results are reported in Figure 4B. We see that in the case of B16 cells irradiated with a dose of 4 Gy, there are 8 times more MN compared to the control. By contrast, the MN

percentages for L929 cells are lower, only 577%, indicating a reduced damage induced by the irradiation in the normal cell line compared to the cancer cell line. Results reporting MN formation after laser-driven electron exposure are scarce. In previous studies, MN formation was reported in normal cells, such as human peripheral blood lymphocytes [40] or MRC5, human fetal lung fibroblasts, but at a much lower level compared to the MN formed following X-ray exposure [42].

4. CONCLUSIONS

This study presents the first implementation at ELI-NP of a 10 PW laser-driven electron irradiation platform applied to normal (L929) and tumoral (B16) cell lines, together with a comprehensive dosimetric characterization.

Multi-detector analysis combining magnetic spectrometry (LANEX), radiochromic film (EBT3), BeOSLs, and online ionization chamber measurements enabled cross-validated dose reconstruction under ultra-high dose-per-pulse conditions. From a dosimetry perspective, the study confirms that plane-parallel ionization chambers such as the Advanced Markus are suitable for online monitoring at low dose-per-pulse conditions; however, under ultra-high dose-rate laser-driven irradiation, significant ion recombination effects must be expected. In the present regime, the uncorrected chamber signal underestimated the absorbed dose relative to radiochromic film. The discrepancy is consistent with general recombination predicted by pulsed-beam theory and highlights the necessity of applying an appropriate recombination correction factor when operating in the FLASH/UHDR domain.

Based on the cross-comparison of charge diagnostics, ionization chamber readings, BeOSL, and radiochromic film measurements, radiochromic film dosimetry was considered the reference method in this study. Unlike ionization chambers, whose response is affected by ion recombination under ultra-high dose-per-pulse conditions, radiochromic films are dose-rate independent and therefore provide a robust determination of absorbed dose in laser-driven irradiation regimes. The agreement between film-based measurements and energy-deposition estimates further supports their suitability as a reliable dosimetric standard for ultrashort, high-intensity electron beams. While online detectors remain essential for real-time monitoring and relative beam stability assessment, absolute dose quantification in the ultra-high dose-rate domain benefits from passive dosimeters that are intrinsically insensitive to instantaneous dose rate effects.

Biologically, ultra-high dose rate laser-driven electrons induced distinct responses in the exposed cells exhibited a dose-dependent reduction in viability and a marked increase in micronucleus formation, consistent with enhanced genomic instability. These findings indicate a cellular response under ultra-high dose-rate electron irradiation. The results establish a reproducible irradiation and dosimetry framework at the 10 PW scale and provide a quantitative basis for further radiobiological investigations at higher doses and with expanded biological endpoints. Although the

results are promising, additional studies are required, incorporating more assays and a broader range of dose exposures.

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